

Blueprint Completo de la Base de Datos — AMS / MagEAM

Versión: 1.0 · 2026-05-12

BD: SQLite (single-file `data/ocp_maintenance.db`)

ORM: SQLAlchemy 2.0 + Pydantic

Modelos: 67 tablas en `api/database/models.py`

Sincronización: auto-create_all en boot (sin Alembic todavía)

1. Vista de las 67 tablas agrupadas

...

CORE — Usuarios + Plantas + Jerarquía

(5 tablas, base de scoping multi-tenant)

users · plants · hierarchy_nodes · criticality_assessments · audit_log

RCM — Análisis de fallas + estrategias

(8 tablas, metodología RCM2/FMECA)

functions · functional_failures · failure_modes · maintenance_tasks

work_packages · fmeca_worksheets · rca_analyses · capa_items

CAPTURE → WR → OT (flujo principal)

(8 tablas, ciclo de vida de la orden)

field_captures · work_requests · managed_work_orders · work_orders

backlog_items · optimized_backlog · planner_recommendations

sync_conflicts

[illegible][illegible][illegible][illegible][illegible][illegible][illegible]

...

2.1 CORE — scoping multi-tenant

[REDACTED]

■ name ■

■ language ■

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

[illegible]

[REDACTED] [REDACTED] [REDACTED] [REDACTED]

```

■ ■ ■ ■ ■ nodes ■ ■ managed_wo ■

```

■ username ■ ■ (PK) ■ ■ parent_id ■ → ■ ■ ... ■

```
■ role ■ ■ plant_id ■ ■ node_type ■ ■ ■
```

■ token_version ■ available ■ code ■ ■ ■

■ plant_id ■

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □

▼

XX

■ assessment_id (PK) ■

■ risk_class (I_LOW.. ■

■ band (B/A/A+/AA) ■

■ status ■

[illegible]

...

▼

XXXXXXXXXXXXXXXXXXXXXXXXXXXX

XXXXXXXXXXXXXXXXXXXXXXXXXXXX

▼

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XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

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■ failure_pattern (A-F) ■

■ wo_id (origen, opcional) ■ ■ source_type (RCA/AUDIT) ■

- method (5why/fishbone) ■ ■ pdca_phase ■
- root_causes (JSON) ■ ■ status (PLAN/DO/CHK/ACT) ■
- recommendations ■ ■ owner_id ■
- approval_status ■ ■ due_date ■

...

2.3 CAPTURE → WR → OT (flujo principal)

...

[illegible]

- field_captures ■
- capture_id (PK) ■
- technician_id ■
- capture_type (VOICE/ ■
■ TEXT/IMAGE) ■
- raw_text ■
- raw_voice_text ■
- images (JSON URLs) ■
- location_hint ■
- gps_lat/lng ■

[illegible]

■

■ source_capture_id (FK)

▼

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

- work_requests ■
- request_id (PK) ■
- source_capture_id (FK) ■
- status: DRAFT → ■
- APROBADO/RECHAZADO ■
- → LINKED ■
- equipment_id/tag ■
- ai_classification ■
- ai_confidence ■
- priority_code ■

■ work_class ■

■ sla_deadline ■

■ created_by ■

■ approver_id ■

■ aviso_coding (M001..) ■

■ planning_group ■

■ area_empresa ■

■ circumstances ■

■ documents (JSON) ■

■ support_equipment ■

■ image_analysis (JSON) ■

■ version (opt. lock) ■

[illegible]

■ work_request_id (string, no FK)

▼

[REDACTED]

■ managed_work_orders ■

[REDACTED]

■ wo_id (PK) ■

■ wo_number (OT-NNNN-NNNN) ■

■ work_request_id (suelto) ■

■ plant_id ■

■ equipment_id / equipment_tag ■

■ technical_location ■

■ wo_type (PM01/PM02/PM03) ■

■ priority_code (P1/P2/P3/P4) ■

■ status: CREADO → LIBERADO → ■

■ PLANIFICADO → EN_PROGRAMACION ■

■ → PROGRAMADO → EN_EJECUCION ■

■ → CERRADO ■

■ is_fast_track (P1/P2 bypass) ■



■ ■ ■ JSON embedded arrays ■ ■ ■

■ operations[] ■

■ materials ■

[illegible]

[illegible]

2.5 RESOURCES

[illegible]

XX

■ values (JSON): MTBF, MTTR, MTBM, availability, etc. ■

[illegible]

■ snapshot_id (PK) · plant_id ■ (DE = Defect ■

■ schedule_adherence ■ events_reported ■

■ reactive_ratio ■ actions_implem. ■

■ ... ■ maturity_level ■

[illegible][illegible]

■ score_id (PK) ■ prediction_id (PK) ■

■ score (0-100) ■ predicted_date ■

■ recommendations ■ failure_mode_id ■

[illegible]

...

2.8 AUDIT + IA + SISTEMA

...

XX

[illegible]

■ timestamp (indexed) ■

■ managed work order, user, settings, ... ■

■ action — CREATE/UPDATE/TRANSITION/APPROVE/... ■

■ user — uuid o "agent:<name>" ■

■ package_id (PK) ■

```
■ wo_ids[] ■
```

■ status (DRAFT/READY/UPLOADED/FAILED) ■

■ format (CSV/JSON) ■

■ generated_by ■

■ created_at ■

[REDACTED]

[illegible]

■ sap_sync_log ■

■ sync_id (PK) ■

■ entity_type/id ■

■ direction (PUSH/PULL) ■

■ status (PENDING/SUCCESS/FAILED) ■

■ payload ■

■ error_message ■

```

    ■ retried_count ■

```

[illegible]

■ import_history ■

■ import_id (PK) ■

■ source_file ■

■ entity_type ■

■ row_count ■

■ errors (JSON) ■

■ imported_by ■

XX

...

2.10 IMPROVEMENT + MOC + RBI

...

[REDACTED]

■ improvement_actions ■ PDCA cycle

■ action_id (PK) ■

■ source_type (RCA/AUDIT/ ■

■ FMECA/USER) ■

Timestamps	`created_at` + `updated_at` (auto-fill por SQLAlchemy `onupdate=datetime.now`)
Soft-delete	NO existe — usar `status: CANCELLED`
Multi-tenant	`plant_id: str` en todas las tablas operacionales
JSON embebido	Para arrays variables (operations, materials, etc.)
Optimistic lock	Campo `version: int` en `managed_work_orders` + WR
Audit trail	`audit_log` tabla universal con `entity_type` + `entity_id`
FK explícitas	Solo donde el borrado cascadeante es deseable (functions → functional_failures → failure_modes)
FK lógicas	Para resiliencia: plant_id, worker_id, material_code (string sin constraint)

4. Índices clave

```
```sql
```

```
-- Lookups frecuentes (todos definidos como index=True en models.py)
```

```
CREATE INDEX idx_audit_log_entity_type ON audit_log(entity_type);
```

```
CREATE INDEX idx_audit_log_entity_id ON audit_log(entity_id);
```

```
CREATE INDEX idx_audit_log_timestamp ON audit_log(timestamp);
```

```
CREATE INDEX idx_wo_status ON managed_work_orders(status);
```

```
CREATE INDEX idx_wo_plant_id ON managed_work_orders(plant_id);
```

```
CREATE INDEX idx_wo_wo_number ON managed_work_orders(wo_number);
```

```
CREATE INDEX idx_wo_priority_code ON managed_work_orders(priority_code);
```

```
CREATE INDEX idx_wr_status ON work_requests(status);
```

```
CREATE INDEX idx_wr_equipment_id ON work_requests(equipment_id);
```

```
CREATE INDEX idx_wr_plant_id ON work_requests(plant_id);
```

```
CREATE INDEX idx_hierarchy_node_type ON hierarchy_nodes(node_type);
```

```
CREATE INDEX idx_hierarchy_tag ON hierarchy_nodes(tag);
```

```
CREATE INDEX idx_hierarchy_plant_id ON hierarchy_nodes(plant_id);
```

```
CREATE INDEX idx_preparativos_wo_id ON preparativos_ot(wo_id);
```

```
CREATE INDEX idx_preparativos_status ON preparativos_ot(status);
```

```
```
```

5. Campos JSON (no relacionales, embedded)

| Tabla | Campo JSON | Estructura típica |
|-----------------------|-----------------------|---|
| `managed_work_orders` | `operations` | `[{seq, description, specialty, hours, completion_pct, actual_hours, task_type}]` |
| `managed_work_orders` | `materials` | `[{code, description, quantity, unit, reservation_code, status, cost_unit}]` |
| `managed_work_orders` | `assigned_workers` | `[{worker_id, name, specialty}]` |
| `managed_work_orders` | `support_equipment` | `[{tag, name, equipment_type, hours, notes, scope}]` |
| `managed_work_orders` | `documents` | `[{name, url, type, uploaded_by, uploaded_at}]` |
| `managed_work_orders` | `description_history` | `[{text, edited_at, edited_by}]`
(append-only) |
| `managed_work_orders` | `post_closure_review` | `{outcome, lessons_learned[], recommendations[]}` |
| `work_requests` | `ai_classification` | `{priority, equipment_match, confidence, suggestions[]}` |
| `work_requests` | `image_analysis` | `{anomalies_detected, component_identified, severity_visual}` |
| `work_requests` | `documents` | `[{name, url, type}]` |
| `audit_log` | `payload` | `{old_value, new_value, context, autonomous?, confirmed_by?}` |
| `weekly_programs` | `wo_ids` | `["wo_id1", "wo_id2", ...]` |
| `field_captures` | `images` | `["url1", "url2", ...]` |

6. State machines

6.1 WorkRequest status

...

DRAFT ■■■■■■ APROBADO ■■■■■■ LINKED (cuando se crea OT)

■ ■

■ ▼

■■■■■■■■ RECHAZADO (sin retorno)

...

6.2 ManagedWorkOrder status (SF-578 cadena)

...

CREADO ■■■■ LIBERADO ■■■■ PLANIFICADO ■■■■ EN_PROGRAMACION ■■■■ PROGRAMADO
■■■■ EN_EJECUCION ■■■■ CERRADO

■ ■

▼ ▼

REPROGRAMADO CANCELADO (terminal)

■

■■■■■■■ PROGRAMADO (loop)

Fast-track P1/P2:

CREADO ■■■direct■■■■ PROGRAMADO (bypass planning)

...

6.3 Preparativos OT status (SF-662)

...

PENDIENTE ■■■■ DESPACHADO ■■■■ EN_TRANSITO ■■■■ RECIBIDO (con conforme=true/false)

■ ■

■■■■■ ANOMALIA ■■■■ DESPACHADO ■■■■ (reintento)

...

7. Compatibilidad SAP PM

Tabla `work_requests` incluye campos pre-mapeados al **Aviso SAP IH01**:

| Campo AMS | Campo SAP | Significado |
|----------------------|---------------|--------------------------------------|
| `aviso_coding` | `QMART` (10) | Tipo de aviso (M001-M005, P001-P003) |
| `notification_type` | `QMNUM tipo` | A1=Aviso Mantto |
| `planning_group` | `INGRP` (10) | Grupo planificación |
| `work_center` | `ARBPL` (20) | Puesto de trabajo |
| `technical_location` | `TPLNR` (100) | Ubicación técnica |
| `area_empresa` | `ABCKZ` (10) | Indicador ABC empresa |
| `reported_by` | `QMNAM` | Autor del aviso |
| `circumstances` | `LTXT` | Texto largo descripción |

Tabla `managed_work_orders` ídem para **OT SAP IW21**:

| Campo AMS | Campo SAP | Significado |
|---------------------|---------------|-----------------------------|
| `wo_number` | `AUFNR` | Número de orden |
| `wo_type` | `AUART` | Clase de orden (PM01/02/03) |
| `priority_code` | `PRIOK` | Prioridad |
| `equipment_id` | `EQUNR` | Número de equipo |
| `planned_start/end` | `GSTRP/GLTRP` | Fechas plan |
| `actual_start/end` | `GSTRI/GLTRI` | Fechas real |

Mapping completo en `sap_mock/data/avisos_campos_blueprint.xlsx` .

8. Operaciones administrativas

Auto-creación (sin Alembic)

```
```python
```

### api/main.py

```
from api.database.connection import Base, engine

Base.metadata.create_all(engine) # crea tablas faltantes al boot

```
```

Backup recomendado

```
```bash
```

### SQLite backup atómico

```
sqlite3 data/ocp_maintenance.db ".backup data/backup_$(date +%Y-%m-%d).db"

```
```

Limpiar audit > 5 años

```
```bash

python scripts/purge_audit_log_5y.py --dry-run # ver qué borraría
python scripts/purge_audit_log_5y.py --confirm --export-csv backup.csv

```
```

Re-indexar RAG (LanceDB separado de SQLite)

```
```bash

python scripts/reindex_rag.py

```
```

9. Tablas que NO existen pero podrían venir (SP8+)

- `cost_centers` (0E tanda) — para opex/capex por CeCo
- `expense_classes` (0E) — clases de gasto SAP
- `failure_symptom_catalog` (SAP-style) — síntomas codificados
- `failure_cause_catalog` (SAP-style) — causas codificadas

- wrench_time_observations — para KPI tradicional
- training_records — horas capacitación
- safety_incidents — integración HSE
- equipment_telemetry — para futuro SCADA real

10. Diferencias work_orders (legacy) vs managed_work_orders

| Aspecto | `work_orders` (Fase 1) | `managed_work_orders` (Fase 4+, actual) |
|-------------------|--------------------------------|---|
| Uso | Legacy, sin lifecycle complejo | OT principal con estado SAP-PM completo |
| Campos | Mínimos (8) | 55+ campos + JSON embedded |
| Lifecycle | Solo created/closed | CREADO→...→CERRADO (7 estados) |
| Auditoría | No | Full audit_log integration |
| Fast-track | No | P1/P2 bypass planning |
| Closure gates | No | 6 pre-close gates |
| **Recomendación** | **Deprecar** en SP9 | Único modelo a usar |

Total: 67 tablas · 562 endpoints · SQLite single-file ~50MB en producción

Generado en commit: `cd83670`